NIKHIL KUMAR

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PROFESSIONAL SUMMARY

Computer Science (Artificial Intelligence) student specializing in Generative AI, LLMs, RAG pipelines, and deep learning systems. Proven ability to architect end-to-end AI solutions - from audio-based CNN classifiers to LangChain-powered retrieval systems - with measurable performance gains. Seeking AI/ML job roles at top-tier technology companies to build production-grade intelligent systems.

EDUCATION

B.Tech in Computer Science with Artificial Intelligence *Jul 2023 – Jul 2027*
UIET, CSJM University, Kanpur ***CGPA: 9.35 / 10.0***

TECHNICAL SKILLS

AI/ML: Machine Learning, Deep Learning, Data Science, Agentic AI, Generative AI, RAG, LLMs, NLP, Prompt Engineering, Feature Engineering, Model Evaluation
Frameworks & Libraries: LangChain, LangGraph, Streamlit, FastAPI, Django, PyTorch, scikit-learn, Librosa, React, Node.js, Express.js
Languages: Python, SQL, C++, JavaScript, HTML/CSS
Databases & Tools: ChromaDB, MySQL, MongoDB, Vector Databases, Git, GitHub, Google Colab, Linux

EXPERIENCE

Artificial Intelligence Intern — SURE ProEd *Mar 2026 – Aug 2026*

- Completed a six-month Generative AI internship covering AI agents, LangChain, LangGraph, tool calling, and Retrieval-Augmented Generation (RAG).
- Designed stateful, multi-step agentic workflows with LangGraph and connected LLMs to external tools through tool calling.
- Strengthened professional skills in structured thinking, workplace communication, accountability, and ethical practice.

Machine Learning Intern — Edunet Foundation *Dec 2025 – Jan 2026*

- Worked on real-world machine learning problem statements, covering data preprocessing, model development, and evaluation.
- Implemented end-to-end ML workflows, including data cleaning, feature engineering, model training, and performance evaluation using industry-standard tools and Python libraries.

PROJECTS

Sound-Based Danger Detection System *Deep Learning · CNN · Librosa · LangChain · LangGraph · LLMs*

- Developed an audio-based danger detection pipeline using Mel spectrograms and frequency-domain features extracted with Librosa, enabling real-time multi-class threat classification.
- Trained a CNN for multi-class danger sound classification, achieving robust performance across varied environmental noise categories.
- Integrated a LangChain/LangGraph-based automated email alert system, triggered on high-confidence danger events to reduce manual monitoring.

Institutional RAG Application *LangChain · ChromaDB · Ollama · Streamlit · Retrieval Systems*

- Architected a fully offline RAG pipeline using LangChain and ChromaDB, eliminating cloud dependency for sensitive institutional data.
- Implemented heading-based semantic chunking and a hybrid retrieval strategy (similarity search combined with MMR), improving context relevance over single-method baselines.
- Added DDGS fallback search for queries outside the knowledge base, improving completeness and user experience.

Next-Day Stock Price Prediction *XGBoost · LSTM · RNN · Python · Feature Engineering*

- Built an ensemble forecasting system combining XGBoost and LSTM for the NIFTY 50 and NIFTY BANK indices, using extensive feature engineering on OHLCV data.

- Achieved R^2 of 0.68 (NIFTY 50) and 0.75 (NIFTY BANK) against LSTM baselines of 0.46 and 0.53, a relative gain of up to 48% in explained variance.

Mental Health Companion Chatbot *LangChain · LangGraph · LLMs · Streamlit · NLP*

- Developed a memory-enabled conversational assistant using LangChain and LangGraph, with multi-step reasoning workflows for contextually accurate, personalized responses.
- Designed a stateful conversation architecture that preserves session context across multi-turn interactions, improving response coherence and user engagement.

Song Recommendation System *Mini-Batch K-Means · DBSCAN · Cosine Similarity · Python*

- Designed an unsupervised music recommendation engine using Mini-Batch K-Means and DBSCAN clustering, with cosine similarity for feature matching.
- Improved the Silhouette Score from 0.12 to 0.26 with Mini-Batch K-Means and reached 0.28 with DBSCAN, a 133% gain over the initial baseline.

CERTIFICATIONS & ACHIEVEMENTS

- Career Essentials in Generative AI – Microsoft & LinkedIn Learning.
- LeetCode 50-Day Streak Badge – consistent problem-solving practice in data structures and algorithms.

LEADERSHIP & EXTRACURRICULARS

- Smart India Hackathon (SIH) 2024 – National-level hackathon participant; collaborated on problem-solving under competitive constraints.
- Core Team Member, ICSCIS International Conference – supported academic event coordination and logistics.
- HackShodh Participant – gained hands-on experience in rapid prototyping and cross-functional teamwork.